Theory Note 8 of 8

Baseline: Why Knowing What Is Normal Matters

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. What is a baseline?

A baseline is a recorded picture of how a network, a system or a user normally behaves over a period of time. It includes things like traffic volume, the protocols and ports in common use, the hosts that usually talk to each other, working hours, the processes and services that normally run, and typical logon patterns.

2. Why it matters for a SOC analyst

- Detection depends on deviation. If you do not know what normal looks like, you cannot tell when something is off.
- It reduces false positives. A backup server sending large amounts of data every night is expected, while the same traffic from a workstation is not.
- It speeds up triage. Comparing an alert with the baseline lets the analyst decide quickly whether it is harmless or needs investigation.
- It helps tune alerts. Thresholds such as failed logons per hour can be set to match the real environment.
- It exposes slow, quiet attacks such as beaconing, gradual data theft and new persistence that signature-based rules often miss.
- It supports incident response by showing when something changed and what is out of place.

3. Examples of normal and deviation

Area	Baseline (normal)	Deviation worth investigating
Network	Mostly DNS, HTTPS and SMB to known servers, with about 200 MB leaving the network per day.	A sudden 20 GB upload at midnight to an unknown IP address.
Logons	Staff log on during working hours, with few failed attempts per day.	Hundreds of 4625 events in minutes, or a type 10 logon at 3 AM.
Processes	Office applications, a browser and the security agent.	winword.exe starting powershell.exe.
Services and accounts	A fixed set of services and administrator accounts.	A new service (7045), or a new admin account (4720 followed by 4732).

4. Building and maintaining a baseline

- Collect data during normal operation for a few weeks, covering flows, logs and an endpoint inventory.
- Record averages and peaks, and allow for business cycles such as month-end, backup windows and patch days.
- Review and update it after changes like new systems, new software or new staff.
- Check that it is trustworthy: a baseline captured while the network was already compromised is not a safe reference.

Key takeaway
A baseline turns "this looks odd" into "this is different from what we normally see". That difference is what lets a SOC analyst make confident decisions quickly.